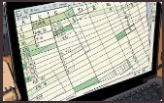


Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to reveal spending patterns, segments, product preferences, and subscription behavior for strategic decisions.



Dataset Snapshot



Scale

3,900 rows $\times$ 18 columns; 37 missing values in Review Rating.



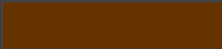
Key Features

Demographics, purchase details, behavior flags (discounts, promo codes), and subscription status.



Summary Stats

Average purchase: \$59.76 · Avg rating: 3.75 · Most common item: Blouse · Majority gender: Male



Data Preparation (Python)



Loading & Exploration

Imported with pandas;
used `df.info()` and
`.describe()` to inspect
structure and
distributions.

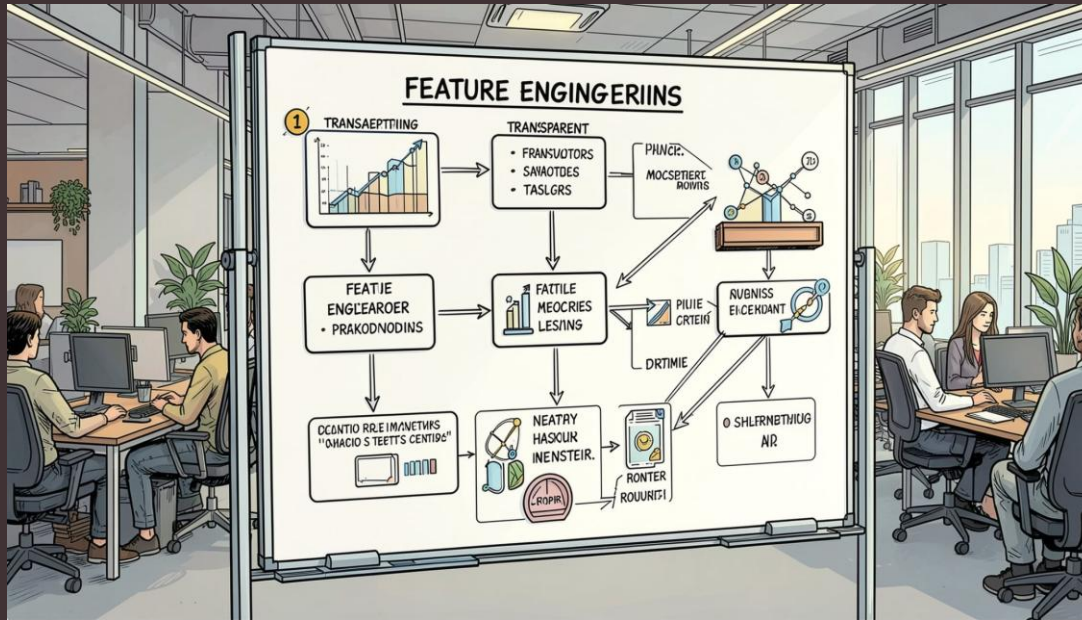
Missing Data

Imputed Review
Rating nulls with
median per product
category.

Standardization

Renamed columns to `snake_case` for readability.

Feature Engineering & Cleanup



- Created age_group bins from Age.
- Derived purchase_frequency_days from timestamps.
- Dropped promo_code_used after checking redundancy with discount_applied.
- Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Revenue by Gender (SQL)



Female

Total revenue: 75,191

Male

Total revenue: 157,890

Men generated the larger share of revenue in this dataset.

Top SQL Insights — Products & Discounts



Top Rated Items

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).



Highest Discount Rates

Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37%.



High-Spending Discount Users

839 rows identified where customers used discounts but spent above average.

Customer Segmentation & Repeat Behavior



Loyal

3,116 customers — largest group.



Returning

701 customers.

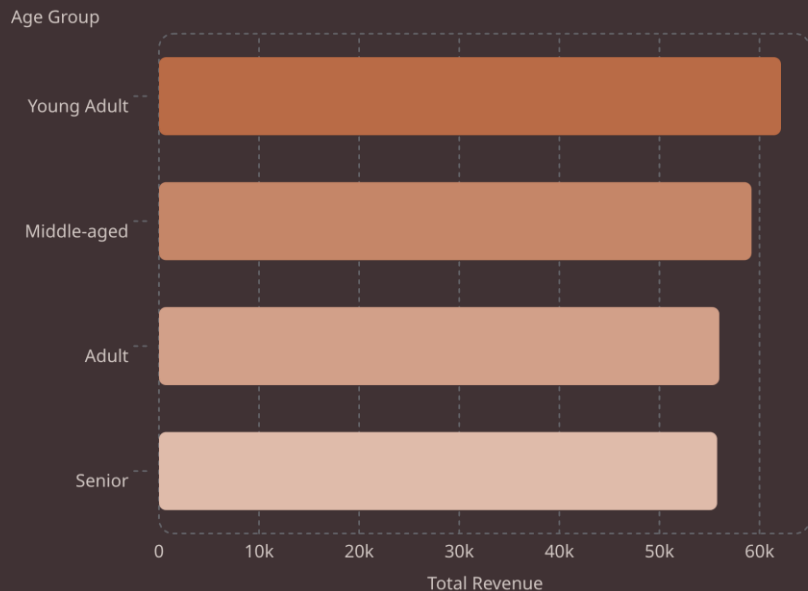


New

83 customers.

Repeat buyers (>5 purchases) show subscription counts: No = 2,518; Yes = 958.

Revenue & Shipping — Age Groups



Shipping comparison: Standard average = 58.46, Express average = 60.48.

Subscribers (Yes): 1,053 customers, avg spend 59.49, total revenue 62,645. No: 2,847 customers, avg 59.87, total revenue 170,436.

Power BI Dashboard

Interactive dashboard summarizing key metrics and visual filters for exploration.

- Number of Customers: 3.9K
- Average Purchase: \$59.76
- Average Rating: 3.75
- Subscription split: 27% Yes / 73% No



The dashboard includes sales & revenue by category, age-group views, and subscription-based segments.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base and recurring revenue.



Loyalty Programs

Reward repeat buyers to move more customers into the Loyal segment.



Review Discount Policy

Adjust discounts to balance sales lift with margin protection—focus on discount-dependent items.



Product Positioning

Feature top-rated and best-selling items in campaigns to boost conversion.



Targeted Marketing

Prioritize high-revenue age groups and express-shipping customers for tailored promotions.

❑ Next steps: operationalize dashboard filters, run A/B tests on subscription offers, and monitor discount impact on margins.